



**HỌC VIỆN
NGÂN HÀNG**
BANKING ACADEMY OF VIETNAM

Web Design

Lecture 6 - Web layout & Responsive design



5.1. Website layout components

5.2. Flexbox

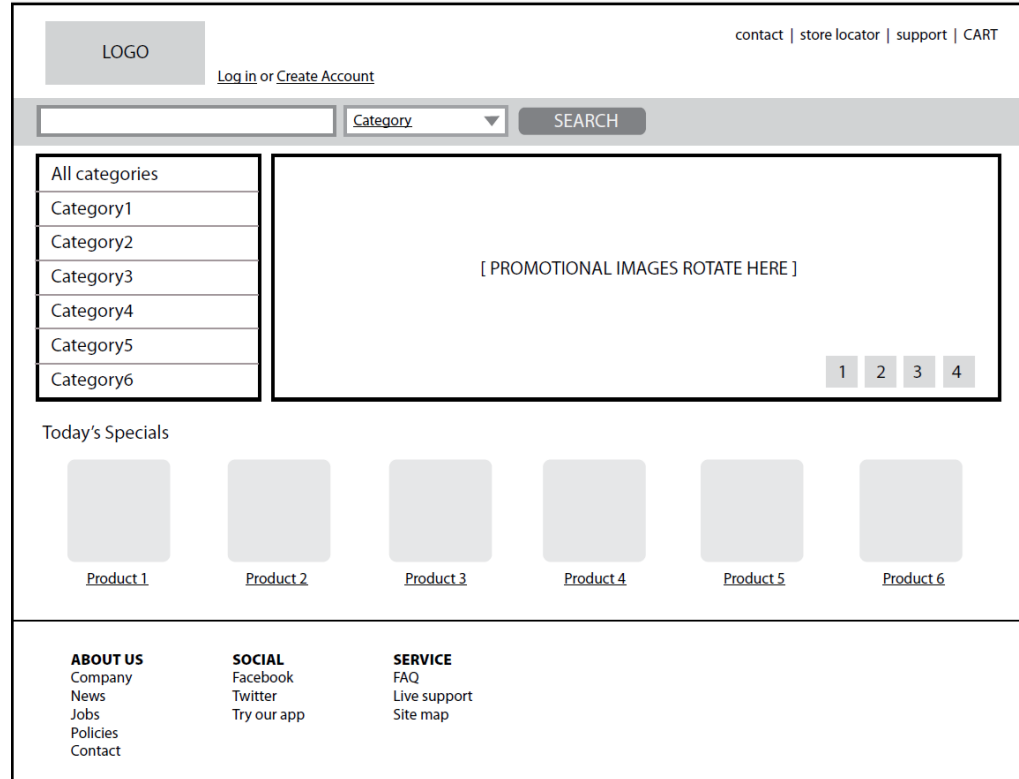
5.3. Grid layout

5.4. Responsive web design

5.1. Website layout components



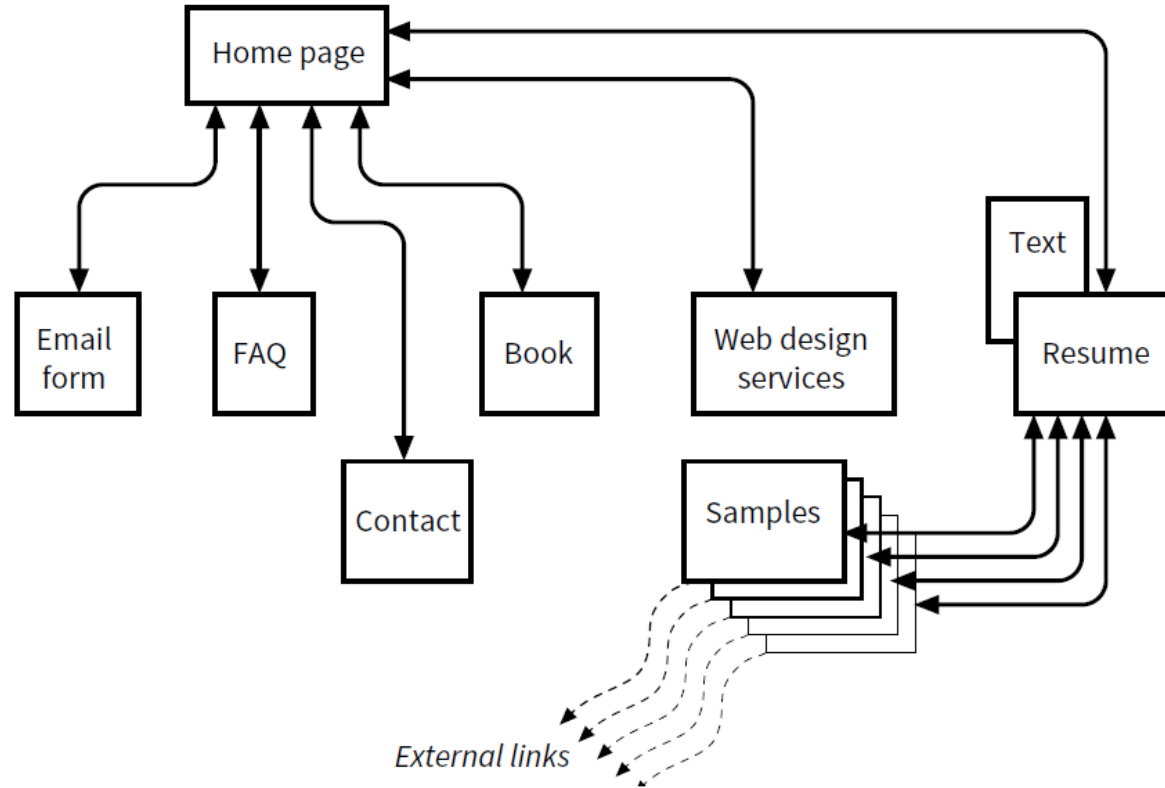
A **wireframe diagram** shows the **structure** of a web page using only **outlines** for each **content type and widget**



5.1. Website layout components



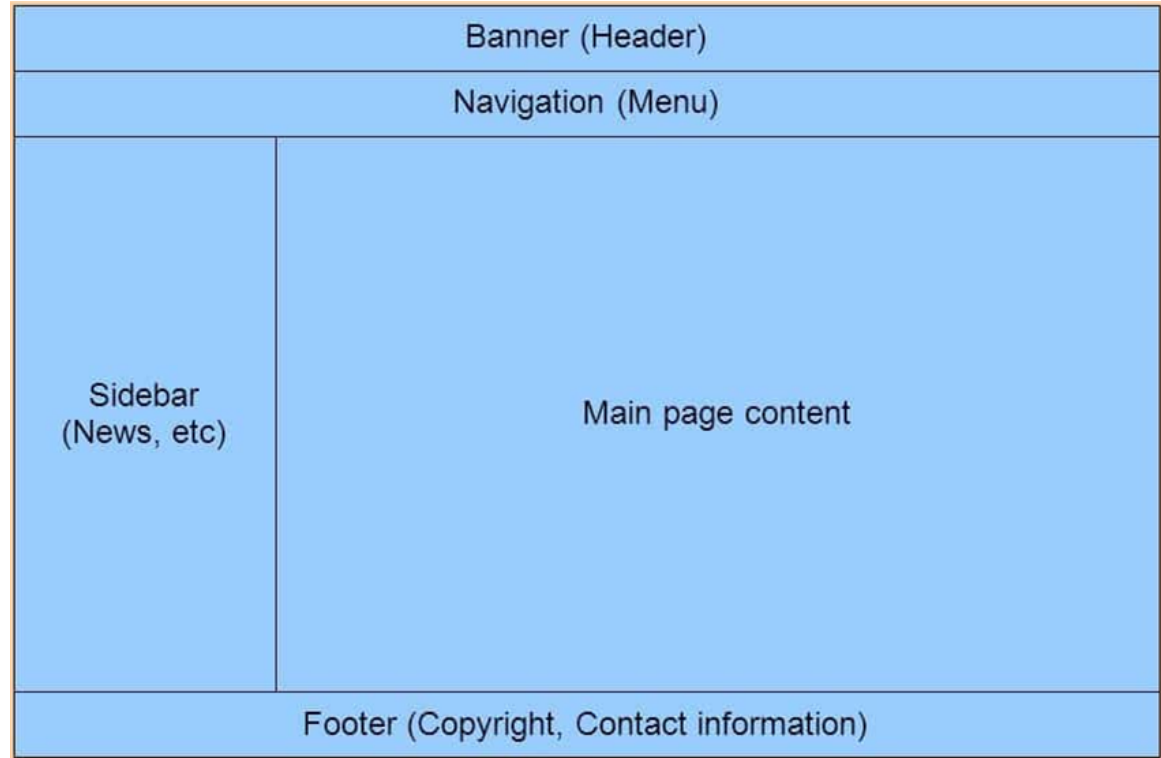
A **site diagram** indicates the **structure of the site as a whole** and how individual pages relate to one another.





Overview:

- Header
- Navigation
- Main content
- Sidebar
- Footer





The top part of a website containing:

- Logo
- Navigation menu (Top menu)
- Banner
- Slogan

The zone at the top of the page that **stays constant and visible as visitors click around your site**





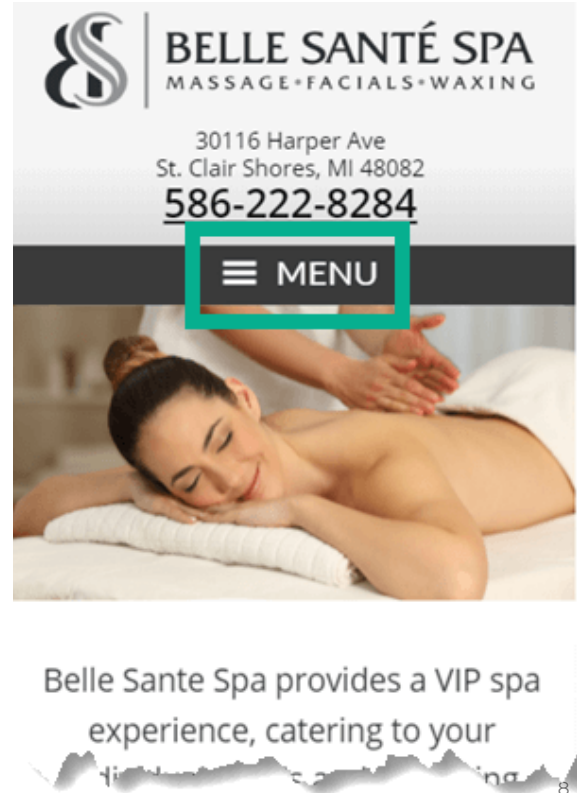
These are the links at the top of the page to help you find what you're looking for. The navigation links are usually ***in the header or just below it.***





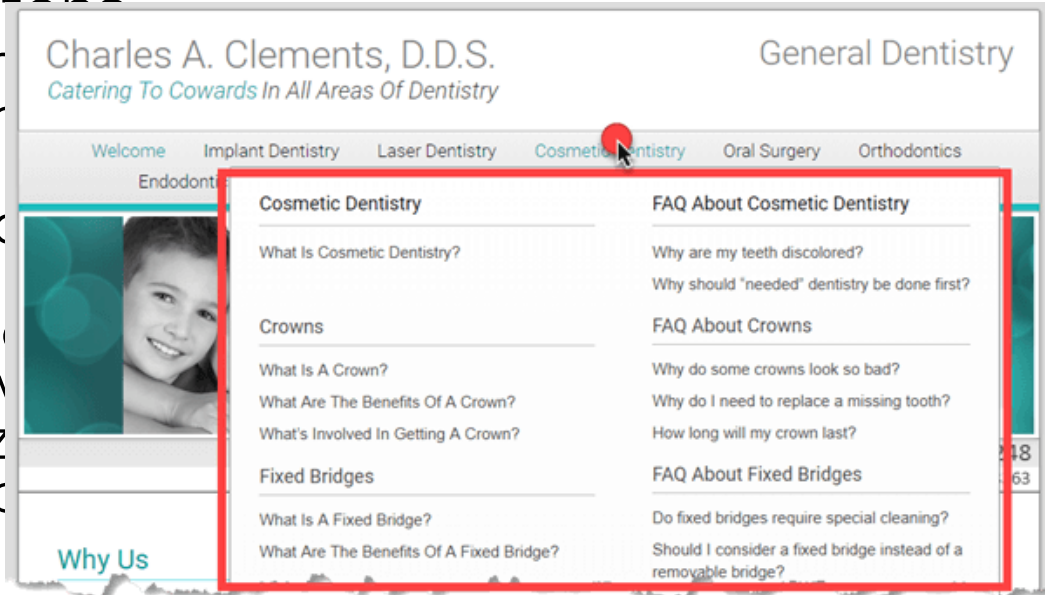
When a site is viewed on a **mobile device**, the navigation usually morphs into an icon with **3 stacked lines** (aka a “*hamburger*” because it looks like the side view of a hamburger) since space is limited on a mobile device.

Tapping the mobile navigation icon usually triggers a vertical or horizontal toggle menu.





On complex sites with thousands of pages, you'll sometimes see a "**mega menu**" which is a *drop-down* menu triggered by hovering over a link in the main navigation. The drop-down mega menu then shows you many links organized into categories and subcategories, kind of like a sitemap.





Website content is the information your visitors consume. Web copy or body copy refers to written text. Website content refers to all the elements used to communicate your message — text, images, video, audio, the whole shebang.





A sidebar is the **narrow vertical column** next or previous to your website content. The sidebar often contains *advertisements, links to other content, calls to action or a search box.*

Think of the sidebar as secondary to your primary website content.





A footer performs the same function as the header — it's a region on a website that's ***constant from page to page*** — except a footer is ***at the bottom of a page*** instead of the top.

You can put whatever you want in a footer. But what's often in the footer is *contact info, privacy policy, terms of use, sitemap, social media icons and links to other important pages on your site.*



I Analyzed 1,100 Medical Websites And Here's What I Learned

Recently my office did a marketing campaign promoting our website design & development services to healthcare providers. Rather than blindly[...]

[Read More >](#)



Does Your Website Really Need HTTPS?

Up until late 2017, I probably would have said don't bother. But times have changed and I now believe it's[...]

[Read More >](#)



How Long Does It Take For A New Website To Appear On Google?

First, a little bad news -- it's impossible to give you an exact timeframe for how long it'll take you[...]

[Read More >](#)

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Search box

Log in / Sign up

Feature images

Slider

Breadcrumbs

CTA: Buttons, pop-ups, ribbons, slide-ins, email optin boxes ... even a simple text link ... are all examples of calls to action. A call to action is a specific and direct request asking your visitor to do something.



Homepage

Product/service page

Item page

About page

Contact page

(mandatory)

Newsletter

Checkout page

Landing page

Blog

404 error page



<https://vnexpress.net/>

<https://dantri.com.vn/>

<https://www.lazada.vn/>

Figure out layout components in each website homepage, category page, item page, etc.



Find top trending web layout in this year

- URL of a sample layout
- What do you like about this
- Work in group

Web design trends 2025

Từ các nguồn trên mạng

	Dark mode	▼		Illustration	▼		Parallax scrolling	▼
	Full-page headers	▼		Micro-interactions	▼		Experimental typography	▼
	Minimalism	▼		Mobile-first design	▼		White space	▼
	3D visuals	▼		Dynamic cursors	▼		Animation	▼
	Color trends 2025	▼		Gradients	▼		Inclusive design	▼
	User experience design	▼		Data visualization	▼		Emerging technologies ...	▼
	AI and personalization	▼		Accessibility and availab...	▼		AI chatbots	▼
	Bold colors	▼		Claymorphism	▼		Grid Design	▼



- Flexbox for greater control over arranging items along one axis
- Grid for honest-to-goodness grid-based layouts, like those print designers have used for decades

“you could create the overall page structure with a grid and use a flexbox to tame the header and navigation elements”



Flexbox allows items to stretch or shrink inside their containers, preventing wasted space and overflow—a real plus for making layouts fit a variety of viewport sizes.

Other advantages:

- The ability to make all neighboring items the same height
- Easy horizontal and vertical centering (curiously elusive with old CSS methods)
- The ability to change the order in which items display, independent of the source



```
<div id="container">
```

```
  <div class="box box1">1</div>
```

```
  <div class="box box2">2</div>
```

```
  <div class="box box3">3</div>
```

```
  <div class="box box4">4</div>
```

```
  <div class="box box5">5</div>
```

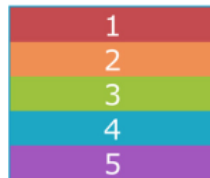
```
</div>
```

```
#container {
```

```
  display: flex;
```

```
}
```

By default, the `div`s display as block elements, stacking up vertically. Turning on flexbox mode makes them line up in a row.



block layout mode

`display: flex;`



flexbox layout mode

flex container





Specifying flow direction

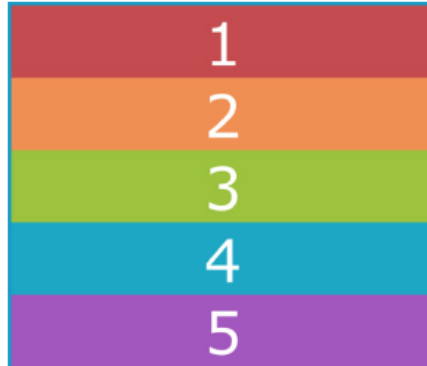
`flex-direction: row;` (default)



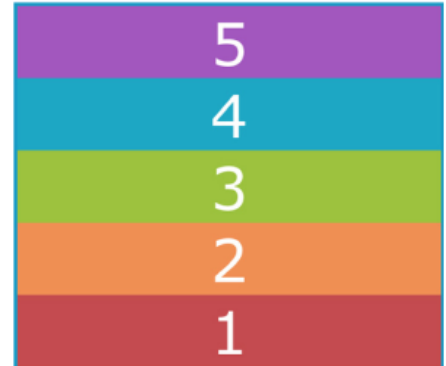
`flex-direction: row-reverse;`



`flex-direction: column;`



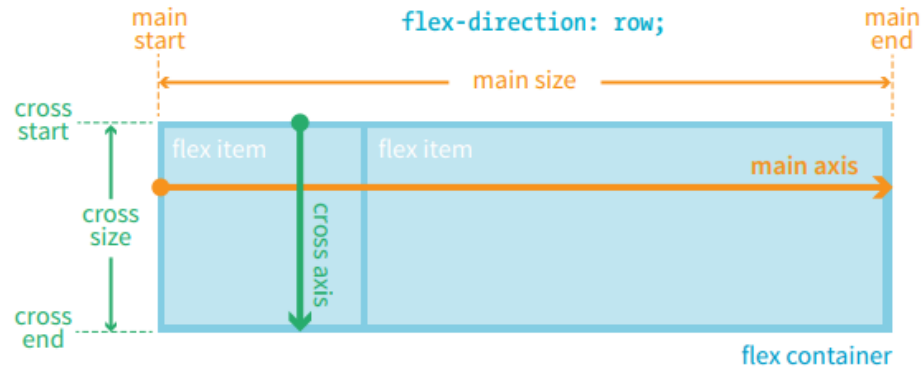
`flex-direction: column-reverse;`



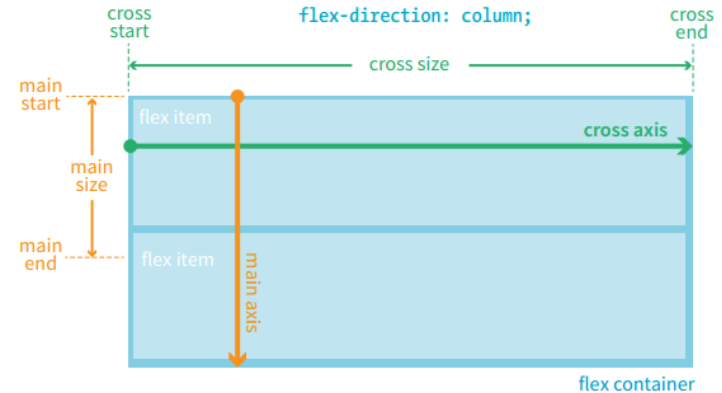


Wrapping onto multiple lines

When `flex-direction` is set to `row`, the main axis is horizontal and the cross axis is vertical.



When `flex-direction` is set to `column`, the main axis is vertical and the cross axis is horizontal.

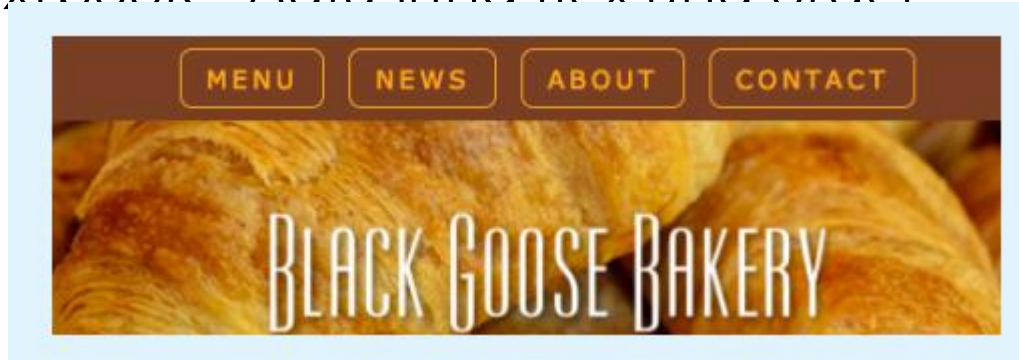




Page 427, textbook - Making a navigation bar with Flexbox

Page 434, textbook - A flexible online menu

Page 445, textbook - Adjusting flex and order





Further reading:

- Controlling the Alignment of Flex Items in the Container, page 428 - 433
- Expanding items (flex-grow), page 437
- Changing the Order of Flex Items, page 442



The CSS Grid Layout Module provides a system for laying out elements in rows and columns (remember that Flexbox lays out elements *on one axis only*)

Chrome 57+, Opera, Firefox 52+, Safari 10+, and iOS Safari 10+ all started supporting the Grid standard free and clear of browser prefixes in March 2017



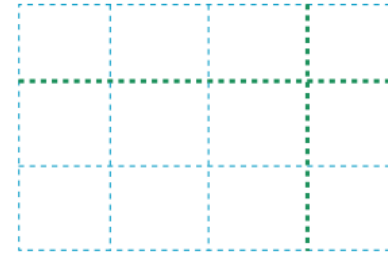
1. Use the **display** property to turn an element into a grid container. The element's children **automatically become grid items**.
2. Set up the **columns** and **rows** for the grid. You can set them up explicitly and/or provide directions for how rows and columns should get created on the fly.
3. Assign each grid item to an **area on the grid**. If you don't assign them explicitly, they flow into the cells sequentially.



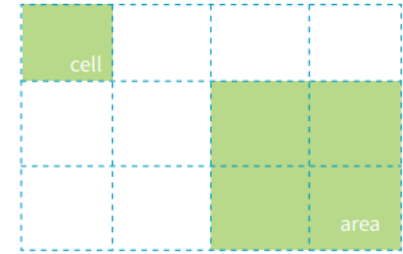
Starting with the markup, the element that has the ***display: grid*** property applied to it becomes the **grid container**

All of its direct child elements automatically become **grid items** that end up positioned in the grid

Grid line ; Grid cell ; Grid area ;
Grid track



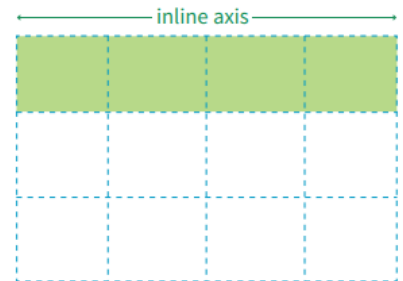
grid lines



grid cell and grid area



grid track (column)



grid track (row)



To turn an element into a grid container, set its ***display*** property to ***grid*** or ***inline-grid***.

In this simple example, the ***#layout div*** becomes a grid container, and each of its children (***#one***, ***#two***, ***#three***, ***#four***, and ***#five***), therefore, is a grid item

THE MARKUP

```
<div id="layout">
  <div id="one">One</div>
  <div id="two">Two</div>
  <div id="three">Three</div>
  <div id="four">Four</div>
  <div id="five">Five</div>
</div>
```

THE STYLES

```
#layout {
  display: grid;
}
```



EXERCISE 16-4. Setting up a grid

Page 461

EXERCISE 16-5. Placing items on a grid

Page 467

EXERCISE 16-6. A grid layout for the bakery page

Page 476



What and Why RWD?

The responsive recipe

Breakpoints

Designing responsively



Responsive web design makes your web page look good on all devices.

Responsive web design uses **only HTML and CSS**. Web pages should not leave out information to fit smaller devices, but rather adapt its content to fit any device.

Responsive web design is not a program or a JavaScript.



Desktop



Tablet



Phone



Since the iPhone appearance in 2007, folks now view the web on phones of all sizes, tablets, “phablets,” touch-enabled laptops, wearables, televisions, video game consoles, refrigerators, ...

In 2016, *mobile internet usage surpassed desktop usage*. That’s where RWD fits in and become important.

In fact, for many web designers, “**responsive design**” is now just “**web design**”



It is called responsive web design when you use CSS and HTML to resize, hide, shrink, enlarge, or move the content to make it look good on any screen.

https://www.w3schools.com/css/tryit.asp?filename=tryresponsive_col-s



One solution was to rely on the ***phone's built-in web display functionality***. By default, mobile devices display an entire web page shrunk down to fit on whatever screen real estate is available. Users can pinch to zoom into the details and scroll around to various parts of the page.

Another approach was to create a ***separate mobile site*** just for small screens and people “on the go.” There are still many companies and services that use dedicated mobile (“*m-dot*”) sites.

→ Far from the “responsive design” we expect



In 2010, Ethan Marcotte gave name to another, more flexible solution in his article “Responsive Web Design”

<https://alistapart.com/article/responsive-web-design/>

- **A flexible grid:** Rather than remaining at a static width, responsive sites use methods that allow them to squeeze and flow into the available browser space.
- **Flexible images:** Images and other embedded media need to be able to scale to fit their containing elements.
- **CSS media queries:** Media queries give us a way to deliver sets of rules only to devices that meet certain criteria, such as width and orientation.
- **Viewport meta:** Makes the width of the web page match the width of the screen



To fit standard websites onto small screens, mobile browsers render the page on a canvas called the **viewport** and then shrink that viewport down to *fit the width of the screen (device width)*





The viewport is the user's visible area of a web page.

The viewport varies with the device, and will be smaller on a mobile phone than on a computer screen.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

open this page on a phone or a tablet

https://www.w3schools.com/css/example_withoutviewport.htm

https://www.w3schools.com/css/example_withviewport.htm



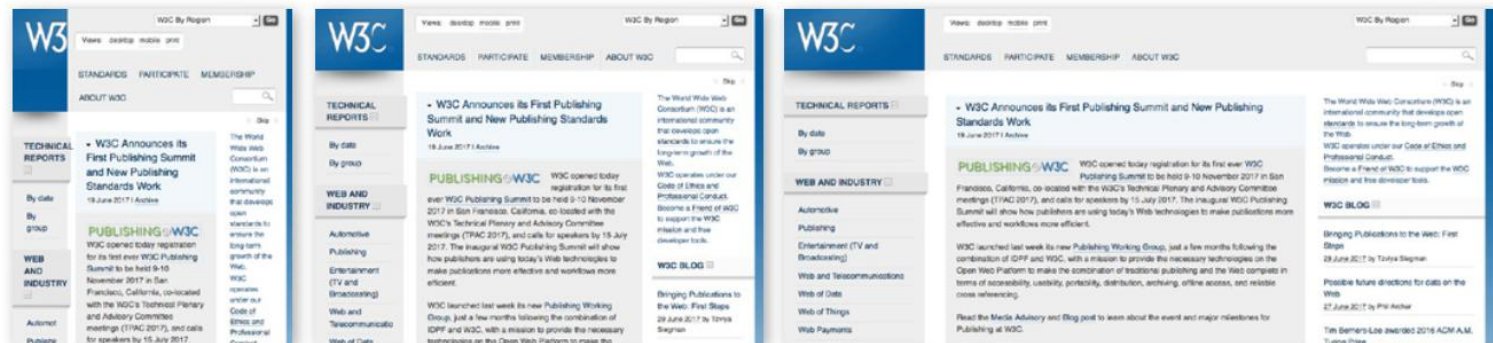
In a fluid layout, the page area and its grid resize proportionally to fill the available width of the screen or window

- ***fr*** or ***minmax()*** units in CSS Grid layout
- ***flex*** property setting in Flexbox
- percentage for old browsers

$$\text{target} \div \text{context} = \text{result}$$

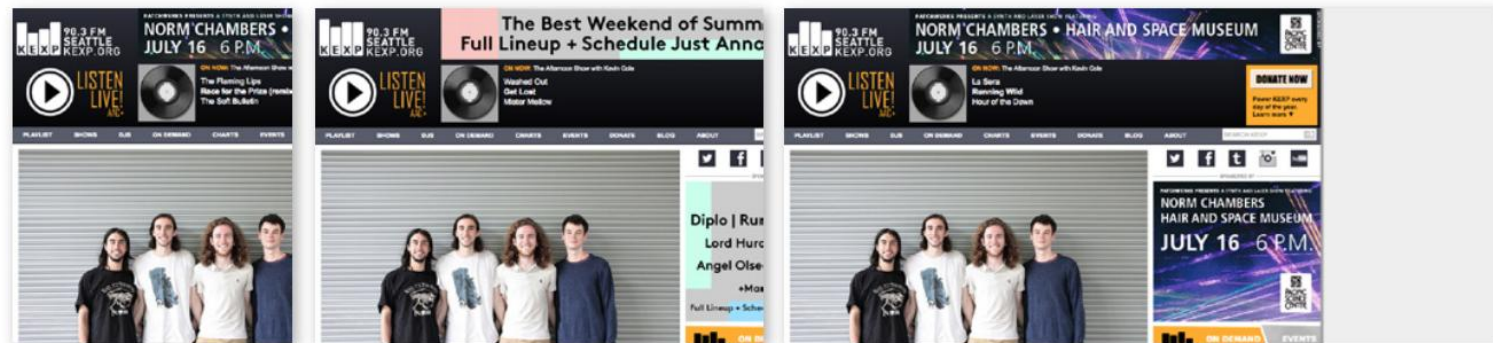
The target is the size of the element you are resizing. The context is the size of the containing element. The result is a percentage you can use in in your style rules.

Fluid layouts fill the viewport proportionally.



w3c.org

Fixed layouts stay the same size and may get cut off or leave extra space.



kexp.org



Hands-on:

https://www.w3schools.com/css/tryresponsive_grid.htm

https://www.w3schools.com/css/tryit.asp?filename=tryresponsive_webpage

https://www.w3schools.com/css/tryit.asp?filename=tryresponsive_cols



The style rule required to make images scale down to fit the size of their container:

```
img { max-width: 100%, }
```

In this book, Philadelphia Ice Creams, comprising the first group, are very palatable, but expensive. In many parts of the country it is quite difficult to get good cream. For that reason, I have given a group of creams, using part milk and part cream, but it must be remembered that it takes smart "juggling" to make ice cream from milk. By far better use condensed milk, with enough water or milk to rinse out the cans.

Ordinary fruit creams may be made with condensed milk at a cost of about fifteen cents a quart, which, of course, is cheaper than ordinary milk and cream.



If pure raw cream is stirred rapidly, it swells and becomes frothy, like the beaten whites of eggs, and is "whipped cream." To prevent this in making Philadelphia Ice Cream, one-half the cream is scalded, and when it is very cold, the remaining half of raw cream is added. This gives the smooth, light and rich consistency which makes these creams so different from others.

The time for freezing varies according to the quality of cream or milk or water; water ices require a longer time than ice creams. It is not well to freeze the mixtures too rapidly; they are apt to be coarse, not smooth, and if they are churned before the mixture is icy cold they will be greasy or

Philadelphia Ice Creams

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But remember: choosing the best image size for performance is part of the responsive design process

https://www.w3schools.com/css/tryit.asp?filename=tryresponsive_image_mediaq



Media queries apply different styles based on characteristics of the browser: its width, whether it is vertically or horizontally oriented, its resolution, and more

They are what make it possible to send a one-column layout to small screens and a multicolumn layout to larger screens on the fly

```
@media type and (feature: value) {  
    /* styles for browsers that meet this criteria */  
}
```

https://www.w3schools.com/css/tryit.asp?filename=tryresponsive_mediaquery



Hands-on:

https://www.w3schools.com/css/tryit.asp?filename=tryresponsive_breakpoints



Page 508